Privacy and Causal Fairness Are Nearly Orthogonal: Graph Surgery Across Three Generator Families

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Abstract

A differentially private synthetic release is published so that nobody has to look at the real records again, which makes the release the only thing anyone can inspect, and makes what it fails to report a problem of its own. We study a fairness intervention inside that setting. DECAF achieves fair synthetic data by factoring a GAN's generator along a causal graph and severing the edges that carry disparity from a protected attribute to an outcome; it has no privacy mechanism, and the technique has never been shown to work anywhere but in DECAF. We port all three of its fairness definitions to nine generators across three unrelated families (marginals-based, GAN-based and diffusion-based, each with differentially private variants) over 2,520 runs at $\varepsilon \in \{1, 10, 1000\}$. Every fairness result is a matched-pairs difference against the run identical in generator, privacy level and seed but with no fairness mechanism, so run-to-run noise cancels inside the pair. Three results. The definitions transfer to every family, though the potency does not: the cut removes 89% of a causal diffusion backbone's disparity and 29% of a marginals-based synthesizer's, because a private synthesizer has no supplied DAG to cut and must bind a structure that noise chose. Privacy and fairness behave as near-independent knobs: the parity-gap reduction at ε=1 is at least as large as at ε=1000 in eight of eleven generator-dataset cells, and in marginals-based methods the cut costs no privacy budget at all, because a severed edge is never measured. And DECAF's own argument that privacy follows for free by post-processing turns out to be sound accounting attached to an unusable output: the drop-in private discriminator leaves the released table with no signal at all (lift over the majority class ≤ 0.001) in 49% of runs, a failure two unrelated changes each repair completely.

1 Introduction

A differentially private synthetic release is an opaque object by construction. The point of publishing it is that the underlying records stay unseen, so every downstream question, including whether the data is useful and whether a model trained on it is biased, must be answered against the release itself or not at all. Two properties of that object therefore matter: what the privacy budget costs, and whether the release tells the truth about itself.

Fairness sits awkwardly inside this picture. The standard intuition is a trade-off: DP noise falls hardest on the smallest groups, so tightening ε should widen disparities and blunt any attempt to correct them. Meanwhile the most attractive fairness mechanism for generated data has no privacy story at all. DECAF [van Breugel et al., 2021] factors a GAN's generator along a causal DAG, so severing an edge at sampling time makes the generator structurally unable to use it, and three fairness definitions become three sets of cuts. It is a single GAN with no privacy guarantee, and nothing about the mechanism has been demonstrated outside it.

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That leaves an awkward dependency. If causal fairness is available only from a generator that cannot offer differential privacy, the two goals are not merely in tension but architecturally incompatible. PreFair [Pujol et al., 2023] shows the incompatibility is not fundamental, embedding one coarse fairness condition into the private structure search of one synthesizer, but stops there: one definition, one algorithm, no evaluation on any other.

We ask whether DECAF's edge surgery is a property of DECAF or of causal factorisation, and what a privacy budget does to it, attaching all three definitions to nine generators from three unrelated families. That includes marginals-based synthesizers, where a severed edge is never privately measured in the first place.

2 Background and related work

DECAF: fairness as graph surgery. DECAF [van Breugel et al., 2021] generates fair synthetic tables from a GAN whose generator is *factored* along a causal DAG: the joint distribution over columns is written as a product of conditionals, $P(X) = \prod_i P(X_i \mid \mathrm{pa}(X_i))$, with $\mathrm{pa}(X_i)$ the columns the graph draws as parents of column X_i. It is the decomposition a marginals-based synthesizer such as MST is also fitting. DECAF gives each column its own sub-network reading only $\mathrm{pa}(X_i)$. The d sub-networks share hidden layers and are trained jointly against a *single* discriminator that scores the assembled row: DECAF's objective is a minimax game between the set $\{G_i\}_{i=1}^{d}$ and one D, so there is no per-column critic. A parent can therefore be severed at generation time by substituting an independently drawn surrogate, leaving the generator structurally unable to use that pathway and requiring no retraining.

Which edges to sever turns a fairness *definition* into an *algorithm*, and DECAF supplies three (Figure 1). **FTU** (fairness through unawareness) cuts only the direct edge $S \to Y$; **DemPar** (demographic parity)[1] cuts every directed path $S \rightsquigarrow Y$; **CF** (conditional fairness) cuts only paths avoiding an *admissible* attribute, sparing disparity routed through a factor deemed legitimate. DECAF's own illustration is the clearest: *Education* $\to$ *Resume* $\to$ *Job* is worth keeping, *Race* $\to$ *Postcode* $\to$ *Loan* is redlining and is not. The machinery is unusually clean and unusually confined: DECAF is a single GAN trained on the standard minimax objective, and it carries no privacy guarantee.

Everything else lives somewhere else. PreFair [Pujol et al., 2023] attacks the other half, adapting *justifiable fairness* to MST [McKenna et al., 2021] by excluding offending edges from candidacy *before* the private structure search runs, so no budget is spent measuring a marginal that will be discarded; it implements one coarse condition, and is neither built nor evaluated for any synthesizer but MST. Causal-DAG diffusion generates column-by-column much as DECAF does but targets fidelity, with no protected attribute and no edge removal [Zhang et al., 2025, Jacob et al., 2026], while private tabular diffusion denoises the row jointly, leaving no per-edge object to cut [Zhu et al., 2024]. Work that does combine privacy with fairness enforces it on a learned *representation* [Hyrup et al., 2025, Sarmin et al., 2025], with an LLM backbone [Nagesh et al., 2025], on a structure learned without a fairness objective [Jia et al., 2025], or *downstream* of the synthesizer [Angelozzi and Arcolezi, 2026]. None exposes an edge.

The gap. Nobody has asked whether DECAF's edge surgery is a property of *DECAF* or of *causal factorisation*. DECAF conjectures the latter twice, calling the method "simple and extendable to other generative methods" and "directly extendable to the more stable WGAN-GP [Gulrajani et al., 2017] and other generative models", but tests neither claim: its own baselines receive protected-attribute removal, never edge surgery. If the conjecture holds, the mechanism should transfer to any generator factored along a graph, including the marginals-based synthesizers that dominate private tabular data [McKenna et al., 2021, Zhang et al., 2014, 2021, McKenna et al., 2022] and including diffusion, the current tabular quality leader and reported superior to GAN and VAE alternatives [Kotelnikov et al., 2023].

[1] DECAF abbreviates demographic parity as "DP". We write **DemPar** throughout and reserve **DP** for differential privacy, because this paper applies both to the same generators and the collision would otherwise be constant.

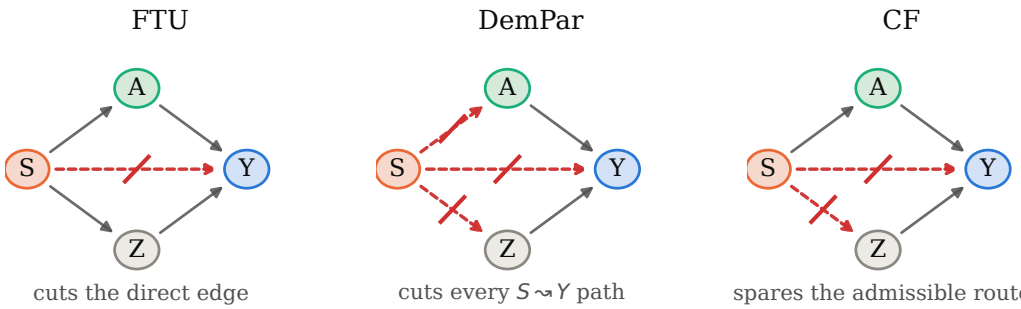


Figure 1: Three fairness definitions as three sets of severed edges, acting on the graph the data is *generated* from, not on the model trained afterwards. S is the protected attribute, Y the outcome, A an *admissible* attribute whose influence is deemed legitimate, and Z a proxy. Dashed red edges are severed. DemPar removes every directed $S \rightsquigarrow Y$ path; CF removes only those that avoid A, so the route $S \to A \to Y$ survives.

3 Porting the mechanisms

We implement FTU, DemPar and CF against one interface and attach it to nine generators from three unrelated families. The definitions do not change; only the object the cut applies to does.

Marginals-based synthesizers (MST, PrivBayes, PrivSyn) measure low-order marginals under differential privacy and fit a graphical model to the noisy counts. A fairness definition becomes a *constraint on structure search*: a filter over candidate marginals, plus a check applied at each selection step. It must be incremental because admissibility is a property of paths, not edges, so whether a candidate creates a forbidden protected-to-outcome path depends on what is already selected. A severed edge is therefore never measured, making the constraint free in the privacy accounting rather than merely cheap. **Causal GANs** (DECAF, plus CTGAN-based [Xu et al., 2019] and DP-SGD [Abadi et al., 2016] variants) use DECAF's surrogate substitution unchanged. **Causal diffusion** keeps the factorisation and replaces each sub-network with a small denoiser trained by supervised noise prediction; the cut is the same code path, because what exposes an edge is that X_i has its own network reading only $\mathrm{pa}(X_i)$, true whether that network is trained adversarially or by denoising.

Protocol. The Adult [Becker and Kohavi, 1996] and COMPAS [Angwin et al., 2016] datasets; nine generators; four mechanisms (NONE/FTU/DemPar/CF); $\varepsilon \in \{1, 10, 1000\}$ at $\delta = 10^{-9}$ for the six generators carrying a differential-privacy guarantee [Dwork et al., 2006]; three protected/admissible role splits per dataset; five seeds; 2,520 completed runs. One Adult split reproduces DECAF's own setup (protected `sex`; admissible `occupation`, `hours-per-week`, `workclass`, `education`), one follows PreFair's Table 1, and one is deliberately strict; COMPAS is absent from DECAF's evaluation, so its splits follow PreFair and two variants. The causal DAG for each dataset is *asserted by hand* from domain knowledge rather than learned from the data — the standard practice in this literature, and the one DECAF, Kusner et al. [2017] and Zhang et al. [2017] all follow. Nothing below depends on that graph being correct, only on it being held fixed across the runs being compared; §5 returns to what the graph's provenance means for an audit.

Metrics. Fidelity is 1-way and 2-way total variation distance to the real table. Utility is the accuracy of an MLP trained on synthetic data and tested on a real held-out 30%, reported as lift over the majority-class rate so that a generator with no signal scores zero. Fairness is that MLP's worst-case demographic-parity gap over the protected attributes, evaluated against real held-out people, which is DECAF's DF-ORIGINAL reference; conditional parity and positive/negative-rate balance are also recorded, and every fairness result below reproduces on all four.

Throughout, **parity-gap reduction** means the amount of that fairness gap a mechanism removes, in gap units, and **fraction removed** means that reduction divided by the gap the same generator left with no mechanism. We keep the two separate because they disagree, and §4 turns on the disagreement. Neither is a statement about fidelity, utility or privacy; those are reported separately and never folded into a single "performance" number. Seed noise rivals the parity-gap reductions themselves (median within-cell SD 0.031 on Adult, 0.092 on COMPAS), so every number below is

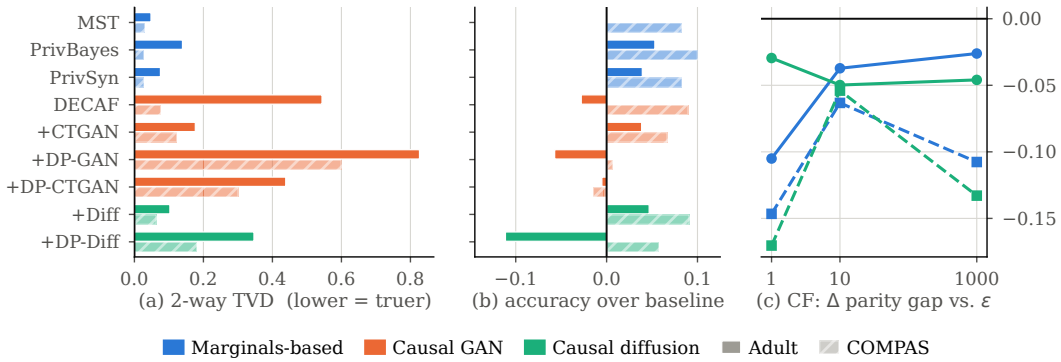


Figure 2: **(a, b)** every generator with no fairness mechanism, averaged over ε, role splits and seeds. A bar at zero in (b) means the data taught a model nothing, and a table with no signal has no disparity left to remove. **(c)** the matched-pairs parity-gap reduction from CF at each privacy level, for the two families present in every cell; more negative is fairer. The reduction does not decay as ε tightens.

Table 1: Matched-pairs fairness change from each mechanism, pooled within generator family. *end* is the gap left behind, Δ gap the mean paired change with a 95% CI, Δ acc the paired accuracy change, *win* the fraction of pairs in which the gap went down. Base gaps are in the family headers.

	Adult				**COMPAS**			
Mech.	end	Δ gap	Δ acc	win	end	Δ gap	Δ acc	win
Marginals-based (MST, PrivBayes, PrivSyn), base gap 0.143 / 0.331, $n = 135$								
FTU	0.121	-0.022 ± 0.020	$+0.002$	33%	0.254	-0.076 ± 0.027	$+0.003$	47%
CF	0.087	-0.056 ± 0.020	-0.005	68%	0.225	-0.106 ± 0.027	-0.011	61%
DemPar	0.071	-0.073 ± 0.021	-0.032	73%	0.230	-0.101 ± 0.029	-0.033	75%
Causal GAN (DECAF, +CTGAN, +DP-GAN, +DP-CTGAN), base gap 0.074 / 0.238, $n = 102/96$								
FTU	0.072	-0.003 ± 0.006	-0.000	33%	0.156	-0.081 ± 0.028	-0.002	66%
CF	0.019	-0.055 ± 0.019	-0.001	44%	0.134	-0.103 ± 0.031	-0.017	60%
DemPar	0.007	-0.068 ± 0.021	-0.003	48%	0.094	-0.145 ± 0.039	-0.049	63%
Causal diffusion (DECAF+Diffusion, +DP-Diffusion), base gap 0.069 / 0.294, $n = 60$								
FTU	0.059	-0.010 ± 0.014	-0.004	52%	0.193	-0.101 ± 0.029	-0.013	85%
CF	0.015	-0.055 ± 0.017	-0.015	72%	0.160	-0.134 ± 0.034	-0.033	80%
DemPar	0.003	-0.066 ± 0.016	-0.007	85%	0.126	-0.167 ± 0.034	-0.088	92%

a **matched-pairs** difference: each run is differenced against the run identical in dataset, generator, ε, role split *and seed* but with no mechanism.

4 Results

4.1 Fidelity and utility

Marginals-based synthesizers are in a different league on fidelity, MST reaching 2-way TVD 0.048 on Adult and 0.030 on COMPAS, which is expected since they optimise almost exactly that quantity; among causal generators, diffusion is strongest (0.103 on Adult against plain DECAF's 0.544) and stays ahead of every GAN variant once DP-SGD is on (0.346 against 0.439 and 0.830). Panel (b) carries the load for what follows: on Adult, MST sits at -0.000 lift, so its headline 0.747 accuracy is exactly the trivial predictor's, and several private causal generators land at or below baseline. A table that teaches a classifier nothing has no disparity to remove and is scored as perfectly fair. Every claim below is a difference *between mechanisms holding the generator fixed*, so this does not confound them; it does mean an absolute fairness number is uninterpretable without the utility beside it.

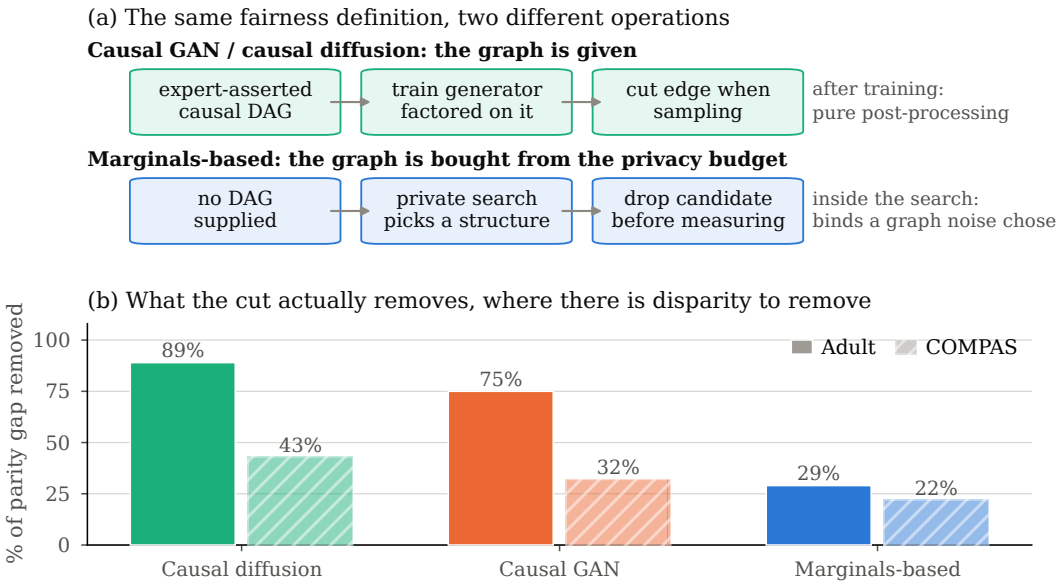


Figure 3: **Why porting is not a port. (a)** The same fairness definition compiles to two different operations. Where a causal DAG is supplied, the cut is applied after training and is pure post-processing; where the graph itself must be bought from the privacy budget, the cut has to be enforced inside the private structure search, against whatever structure the noise could afford. **(b)** The consequence, as the fraction of the parity gap that conditional fairness removes, over matched pairs with disparity to remove and a model that learned something. The definition transfers everywhere; the potency does not.

4.2 Fairness: the definitions transfer, the potency does not

The parity-gap reduction is portable in direction and significance. In every family, on both datasets, at every privacy level, cutting edges reduces the downstream parity gap, and the reduction is many times its own standard error. DECAF's conjecture is correct in the sense that matters first: nothing about the three definitions requires a GAN.

It is not portable in magnitude, and the raw averages hide that. Pooled over Adult, CF removes 0.056 ± 0.020 of gap from the marginals family, 0.055 ± 0.019 from causal GANs and 0.055 ± 0.017 from causal diffusion. Those three means agree to 0.002, and we initially read that as invariance. It is an averaging artifact. The nine per-generator reductions on Adult span 0.002 (MST) to 0.173 (DECAF), and each family happens to contain both a generator that reproduces the disparity and one that collapses to no signal, so the family means are pulled to a common value by composition rather than by any shared property of the mechanism. The fraction removed asks the scale-free question instead: of whatever disparity a generator actually produced, how much does the cut take away? Restricted to matched pairs where there is disparity to remove (baseline gap > 0.05) and the generator learned something (lift > 0.02), CF removes 89% of the gap for causal diffusion and 75% for causal GANs, against 29% for the marginals-based synthesizers (Adult; 43%, 32% and 22% on COMPAS), plotted in Figure 3b. The family term explains none of the variance in the raw parity-gap reduction ($\eta^2 = 0.000$) and a quarter of it in the fraction removed ($\eta^2 = 0.25$).

The gap has a structural cause: it is not one mechanism but two (Figure 3a). DECAF is handed a causal DAG and severs edges by surrogate substitution at sampling time, after training, so the cut is pure post-processing on a graph somebody supplied. A private marginals-based synthesizer is handed no DAG; it buys a structure out of the privacy budget, so the same definition has to be enforced upstream, inside structure selection, where it binds a graph that noise chose rather than one a domain expert supplied. The definitions are portable because they are stated over graphs. The potency is not, because the two families do not have the same graph to operate on. This is why porting the mechanism is a research question rather than an engineering exercise: the second implementation is not DECAF's mechanism moved to a new backbone, it is a different operation that satisfies the same definition.

Table 2: **Where to intervene, on identical data.** AIM synthetic training data, real held-out test set, XGBoost, 5 seeds, run inside the unmodified benchmark harness of Angelozzi and Arcolezi [2026]. Entries are |parity gap| / accuracy; lower gap and higher accuracy are better. The eight mitigators act on the released table or on the model trained from it; our three act inside the generator. Grey marks a cell that taught the classifier nothing — accuracy at or below the majority-class rate (0.755 Adult, 0.534 COMPAS) or a degenerate constant predictor (zero recall) — so its parity gap is not interpretable. `n/a`: the mitigator returned an undefined metric in every seed.

		Adult		COMPAS	
stage	intervention	ε=1	ε=10	ε=1	ε=10
	no mitigation	0.196/0.824	0.195/0.831	0.224/0.636	0.228/0.655
pre	Reweighing	0.106/0.820	0.095/0.826	0.137/0.637	0.099/0.648
pre	Disp. impact remover	0.173/0.807	0.181/0.824	0.186/0.647	0.193/0.644
pre	Learned fair repr.	0.000/0.755	0.000/0.755	0.019/0.503	0.030/0.507
in	Exp. gradient	0.117/0.818	0.110/0.825	0.141/0.635	0.071/0.638
in	Grid search	0.205/0.821	0.185/0.829	0.224/0.636	0.228/0.655
post	Reject option	0.048/0.744	0.040/0.752	0.041/0.516	0.041/0.568
post	Equalised odds	0.073/0.794	0.077/0.798	0.021/0.597	0.030/0.620
post	Calib. eq. odds	0.057/0.793	0.064/0.798	0.666/0.556	0.642/0.563
in-gen	FTU	0.187/0.823	0.185/0.824	0.188/0.632	0.212/0.649
in-gen	DemPar	0.020/0.767	0.010/0.747	0.193/0.639	0.175/0.637
in-gen	CF	0.189/0.823	0.183/0.822	0.176/0.632	0.224/0.656

That constraint is free in the privacy accounting — a severed edge is never measured, so no ε is spent on it — but the freed budget does not buy the release anything back. Across the marginals-based generators, matched-pair 2-way TVD is *worse* under the cut, by 0.008 for DemPar, 0.003 for CF and 0.001 for FTU, ordered exactly by how much structure each removes.

The reason is visible in the algorithm. MST spends its structure budget over a fixed number of rounds, one per edge a spanning tree needs, and sets the per-round ε from that count *before* selection begins — so it does not depend on how many candidates survive the fairness filter. Excluding edges therefore leaves no more budget for the edges that remain. It only shrinks the pool each round draws from, and the exponential mechanism settles for a lower-scoring edge. Prune hard enough and a round has no admissible candidate at all: selection stops early and the model is a spanning *forest* rather than a tree, which is strictly less structure for the same ε. Free in the privacy accounting is not free in fidelity.

The predicted ordering falls out of the data. Ranked by parity gap left behind, NONE $>$ FTU $>$ CF $>$ DemPar holds in five of six (dataset $\times$ family) cells; in the sixth, marginals on COMPAS, CF and DemPar swap by 0.005, far inside either interval. CF landing between the other two is a substantive check, not a formality: it is *designed* to spare admissible disparity, so wherever a role split gives DemPar a mediator to cut and CF one to spare, CF must recover strictly less, and wherever the two separate by more than noise, it does.

Fairness through unawareness is weak, and on Adult barely acts. Pooled over Adult, FTU moves the gap by -0.013; the distribution is the informative part, leaving it exactly unchanged in 41% of pairs, better in 37%, worse in 23%. The unchanged cases have a concrete cause: for MST the mechanism is a no-op in all 45 pairs, because the direct protected-to-income marginal is one the private structure search never buys, so there is nothing to exclude. Where the cut bites, its sign is near a coin flip, the influence rerouting through proxies. This is the textbook criticism of unawareness, confirmed constructively: the cheap cut fails where the two path-based cuts, on the same generators and seeds, work.

The utility cost tracks the aggressiveness of the cut and is modest for the cut worth making: pooled over Adult, DemPar costs 1.7 accuracy points and CF 0.6; on COMPAS, 5.0 and 1.7. CF stays under two points in five of six cells.

4.3 How this compares to intervening downstream

Everything above measures the mechanism against itself. To place it against the alternatives an institution actually has, we ran the benchmark of Angelozzi and Arcolezi [2026], which asks where in a differentially private synthetic pipeline a fairness intervention should go and answers with three downstream stages — repairing the released table, constraining the classifier, and post-processing its predictions — and added ours as a fourth. Their harness is used unmodified: their preprocessed data, their splits, their classifier and their aif360 metrics. The only change is that our arm's training table is generated by our synthesizer under a mechanism. We report the AIM arm, AIM being the synthesizer here whose data clears the majority-class rate on both datasets.

The in-generator cut is competitive exactly where the generator encoded the disparity in the first place. On Adult at $\varepsilon{=}1$, DemPar removes 89% of the parity gap that the same generator leaves with no fairness mechanism, 0.187 to 0.020 — the largest reduction of any interpretable intervention in the panel. The best downstream alternative, calibrated equalised odds, reaches 0.057 while keeping 2.6 more points of accuracy, so the two sit on one frontier rather than either dominating. On COMPAS the cut is weak: 0.212 to 0.175 at $\varepsilon{=}10$, against 0.071 for the exponentiated-gradient reduction at the same accuracy. The reason is §4's finding seen from outside. The privately selected structure barely encodes the protected-to-outcome dependence, so there is little for a cut to sever, whereas a mitigator acting on the trained model is not subject to that limit. (At $\varepsilon{=}10$ on Adult our own cut overshoots into the degenerate region of §4, which is why that cell is greyed.)

We therefore do not claim the in-generator cut dominates downstream mitigation; on this evidence it does not. What survives the comparison is not the size of the reduction but who can guarantee it happened. A downstream mitigator is applied by whoever trains a model, and a published table has consumers the publisher will never meet, whereas a cut inside the generator ships with the table.

4.4 Privacy is close to an orthogonal axis

The folk expectation is a trade-off: DP noise falls hardest on the smallest groups, so tightening ε ought to widen gaps and blunt any correction. We do not see it. The mechanism removes at least as much gap at the defensible $\varepsilon{=}1$ as at the nominal $\varepsilon{=}1000$ in nine of twelve (family $\times$ dataset $\times$ {CF, DemPar}) cells, and two of the three exceptions are COMPAS GAN cells where the backbone has already collapsed at $\varepsilon{=}1$ and has no disparity left to remove. For marginals methods on Adult, CF removes 0.105 against 0.026, and on COMPAS 0.147 against 0.108. Part of that is DP noise inflating the base gap and so leaving more to remove, but the relative reduction moves the same way (68% against 18% on Adult). For causal diffusion the parity-gap reduction is flat: on Adult, CF removes 0.030, 0.050 and 0.046 across the three privacy levels, intervals overlapping. One deviation we cannot explain is visible in Figure 2c, the reduction at $\varepsilon{=}10$ being weaker than at either neighbour in both families and both datasets; we report it rather than smooth it. Fidelity does pay for privacy, but least where it matters: for marginals methods, $\varepsilon{=}1000 \to 1$ costs 2-way TVD only $0.045 \to 0.137$ on Adult and $0.025 \to 0.035$ on COMPAS.

One private generator fails outright, on a claim DECAF makes. DECAF argues from the post-processing theorem that it "can be directly used for generating private synthetic data by replacing the standard discriminator by a differentially private discriminator." The accounting is correct; the output is not. Doing exactly that collapses the synthetic outcome to a single class in 168 of 360 runs (47%), every one of them on COMPAS at $\varepsilon{=}1$ included. Two independent changes each remove the failure: a CTGAN representation under the same adversarial training ($1/360$), and DECAF's representation with diffusion in place of the critic ($0/360$). Two unrelated repairs locate the cause in DP-SGD on a weight-clipped critic, not in private causal generation.

5 Discussion

Post-processing is an accounting argument, not a guarantee that anything comes out. DECAF states that it "can be directly used for generating private synthetic data by replacing the standard discriminator by a differentially private discriminator," citing the post-processing theorem. The privacy claim is correct and we do not dispute it. What our runs show is that the sentence is nonetheless the wrong advice: done literally, it produces a table carrying no downstream signal in 49% of runs across the two DP-GAN backbones, rising to 80% for the DP-CTGAN backbone on COMPAS at $\varepsilon{=}1$. The

failure is narrow and locatable. Keeping adversarial training but changing the data representation fixes it, and keeping the representation but dropping adversarial training for diffusion fixes it; two unrelated repairs of one symptom put the cause in DP-SGD applied to a weight-clipped critic. The general lesson is worth stating plainly, because this style of argument is common: post-processing composes privacy budgets, and says nothing whatever about whether the composed system still generates data. Papers that reach for it owe an experiment.

The fairness constraint is free in the accounting, but not in fidelity. In the marginals-based family a severed edge is excluded from candidacy before the private structure search runs, so no ε is ever spent measuring it; in the DECAF-style families the cut happens after training and is itself post-processing. Both are free in the privacy accounting, for different reasons, and against the standard picture in which fairness is a tax paid out of the privacy budget that is a strong negative result. It is worth being precise about what the freed budget does *not* do, though: it does not buy a better-fitting release. Across the marginals-based generators, matched-pair 2-way TVD is slightly worse under the cut, by 0.008 for DemPar, 0.003 for CF and 0.001 for FTU — ordered by how much structure each removes. The reason is structural rather than accounting: MST fixes its per-round ε from the number of edges a spanning tree needs, before selection begins, so excluding candidates leaves no extra budget for the edges that remain and merely shrinks the pool each round draws from. Prune hard enough and a round has no admissible candidate at all, so the model ends as a spanning forest rather than a tree — less structure for the same ε. Fairness costs no privacy here; it still costs a little fidelity.

Opacity has a measurable direction, and it flatters. DECAF's Definition 4 warns that a distribution can look fair scored against itself and still yield disparate impact under the distribution the model is deployed into. We can put a number on it. Every fairness result above is measured against real held-out people; an analyst holding only the release must score the model against the synthetic data instead. Under DemPar on Adult, the release reports a parity gap of 0.010 while the same models sit at 0.035 against real people, a $3.5\times$ understatement. The direction is the uncomfortable part: the distortion is *largest* for the mechanism that most reduces the true gap, because the intervention acts on the generative process and so is fully realised in the synthetic distribution, while the deployed model meets a world still containing every proxy the generator suppressed. A private release used as its own audit substrate will certify models it should not, and the better the fairness mechanism, the more confidently it will do so. Reporting both references costs one extra evaluation on data the publisher already holds.

A tight privacy budget manufactures fairness that is not there. On Adult at $\varepsilon{=}10$, both private GAN backbones report a parity gap of 0.000 before any mechanism is applied, with lift at or below 0.001 over the majority class. A table that teaches a classifier nothing contains no disparity to find. Any privacy-fairness benchmark that reports gaps without utility beside them is unfalsifiable in exactly the regime it exists to study.

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